Effect of Food-Away-From-Home Tax Rate on Grocery Spending in the United States: Evidence from 2020–2024

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Abstract

Food tax rate is a crucial factor when households determine the allocation of budgets. In this paper, we analyze the impact of Food-away-from-home (FAFH) tax rate on household grocery food expenditure. We use a two-way-fixed-effect model to capture the impact through data of 71,844 households in Current Population Survey (CPS) from 2020-2024, combining the constructed state-level tax data from United States Department of Agriculture (USDA)'s food tax database. The model yields a negative and statistically insignificant estimation on the effect of FAFH tax rate, whereas the Food-at-Home (FAH) tax rate is positively and significantly associated with grocery spending. The results carry implications for progressive food tax policy design: FAFH taxation is a relatively efficient and distributionally neutral revenue instrument. Key limitations of the analysis include cognitive bias in self-reported expenditure, measurement error in the constructed tax rate variable arising from the aggregation of subsidy-exempt Supplemental Nutrition Assistance Program (SNAP) transactions, and the in-

ability to resolve the endogeneity of SNAP participation using available instrumental variables.

Keywords: Food-Away-From-Home Taxes, Grocery Expenditure, Food Consumption

1. Introduction

Food is a fundamental component of social well-being and quality of life, which makes food tax policy a long-examined subject in public policy analysis due to its direct effect on food prices. The United States Department of Agriculture (USDA) classifies food taxes into two primary categories: Food-Away-From-Home (FAFH) and Food-At-Home (FAH). Food-Away-From-Home refers to meals consumed outside the home, such as those purchased at restaurants and cafeterias, while Food-At-Home encompasses food sold at grocery stores and markets. These two categories influence food consumption in distinct ways. Higher grocery taxes create incentives for households to dine out, which can in turn lead to reduced caloric intake or nutritional imbalance. This dynamic stands in contrast to the circumstances of lower-income households, who are often compelled to either reduce spending on non-essential items or struggle to meet their basic needs, depending on their level of income. Because lower-income households allocate a disproportionately large share of their income to food expenditure relative to middle- and high-income households, they bear a heavier burden when grocery taxes are imposed (Zheng et al., 2021). This paper examines the impact of FAFH taxes on FAH consumption, as measured by weekly grocery expenditures.

The paper begins with an introduction to the research topic and the central research question. It then reviews key findings from the existing literature, including the moderating role of SNAP participation, the substitution effect between FAH and FAFH, and the endogeneity challenges inherent in this type of analysis. The data section that follows describes the sources used for model estimation and outlines the characteristics of the sample population, as well as how those characteristics inform the empirical strategy. Subsequently, the paper presents the baseline model specification, followed by a discussion of the main results, their empirical implications, and the study's limitations. The paper concludes with a summary of the findings.

2. Literature Review

Plenty of research has examined the impact of taxes on food expenditures. A systematic review of relevant studies suggests that most research found taxation on the targeted categories of food would decrease its sales (Andreyeva et al., 2022). For instance, Agarwal et al. tested the impact of a policy introducing taxes on fast foods and found 3.9% decreases on fast food sales (Agarwal et al., 2023). However, the effect on food expenditure depends on tax design, as well as population characteristics, which is illustrated in the following paragraphs.

2.1 SNAP Participation Effect

Supplemental Nutrition Assistance Program (SNAP) participation is likely to discard the taxation effect, as the SNAP recipients are exempted from state and local level taxes. The research of Dong and Stewart (2021), Zheng et al. (2019), and Wang et al. (2023) all found a differentiated effect of taxes on SNAP recipients and other household groups in their studies. Dong and Stewart (2021) assessed the association between the type of tax on FAH spending on the three groups i.e., SNAP participants, SNAP non-recipients who are eligible for it, and non-eligible households (mostly are high-income households), utilizing the country-level sales tax data and 2012-13 National Household Food Acquisition and Purchase Survey (FoodAPS) data. The research found that the eligible non-participants of SNAP reduce FAH spending as grocery tax increases, while the other two groups are less sensitive to grocery taxes. Besides, SNAP households make more FAH spending with higher restaurant taxes. (Dong & Stewart, 2021)

2.2 Substitution Effect Between FAFH and FAH

Some studies suggest some level of substitute effect between FAH and FAFH. However, literature suggests mixed results when examining them using real world data. Zhao et al. found that with grocery taxes levied, FAH spending reduces, and households rely more on food in FAFH category. Whereas restaurant taxes will lead households to spend more on grocery items (Zhao, et al. 2025). Zheng et al. (2019) examined how expenditure on grocery and restaurant food is affected by a food sales tax, and if SNAP participants are less sensitive to food sales tax, using county-level sales tax data with the one-year (2012-13) USDA's FoodAPS data. They found that grocery tax reduces expenditure on grocery items and increases restaurant food expenditure, whereas a sales tax on restaurant food incentivizes people to increase their grocery food expenditure. With respect to FAFH spending, a 1-percent increase in the grocery tax led to a $1.46 increase in weekly FAFH spending, while SNAP participation decreased the spending. Restaurant tax does not have a statistically significant relationship with FAFH spending (Zheng, et al., 2019). However, some researchers suggest contradictory evidence. Richards and Mancino found that cross-price elasticities of demand are small, which indicates that consumers are unwilling to substitute between food-at-home and any type of FAFH; or among types of FAFH (Richards & Mancino, 2014).

2.3 Endogeneity

Although not the majority, some research raises concerns about the endogeneity problem of multiple variables in the study. In the study of Zheng et al., they mention that SNAP participation might be affected by a self-selecting effect of individuals, which makes this variable correlated with the unobserved problem. They relied on three instrumental variables to address the problem of endogeneity of SNAP participation, which captured the variation in SNAP eligibility requirements and administrative options: the use of biometric technology,

adoption of simplified reporting, and SNAP outreach spending. (Zheng et al., 2019). In another study examining the effect of grocery tax rates on food insecurity, Zheng et al. encountered the challenge of spatial endogeneity on grocery tax. The authors picked a variety of instrumental variables and located the types of local governing authority (Home rule and the Dillon rule) as a strong candidate for instrumental variable for grocery tax, as the type governs whether local governments can alter sales taxes or sales tax exemptions. They then performed a Wald test on the instrument variables, which suggests that the taxes are not endogenous and thus has to treat it as exogenous in the study (Zheng et al., 2021).

Overall, as there is no determining result on the substitute effect between tax rates and expenditure on different food categories, we aim to address the effect of FAFH taxes on grocery food expenditure, while controlling the variable of SNAP participation, demographic characteristics, as well as the FAH tax rate to control for the substitute effect. Based on the main findings from previous research, we hypothesize that an increase in FAFH tax rate would result in a decrease in food expenditure overall.

3. Data

The analysis draws on household-level data drawn from the Current Population Survey (CPS) food security database and the United States Department of Agriculture (USDA) database, spanning 2020 to 2024. Socio-economic characteristics are retrieved from the CPS database for each respondent, including socio-economic characteristics, including the year of interview, state code, age, sex, marital status, race, employment status, disability, citizenship, number of children, educational attainment (college degree or no college degree), household income relative to the 185% federal poverty level, metropolitan residence, permanent address, and SNAP benefit receipt in the year of interview (Current Population Survey, 2024). Since respondents are subject to the food tax policy of their state of residence, state-level FAH and FAFH tax rates are calculated based on per-capita food sales values in each category

before and after taxes, using data from the USDA food tax database. These rates are then matched to each observation according to their state code and year of interview. We calculate the FAH and FAFH tax rates by the equation following:

$$\tau FAFH_{st} = \frac{(FAFH(\text{spending with taxes per capita})_{st} - FAFH(\text{spending without taxes per capita}_{st}))}{FAFH(\text{spending without taxes per capita}_{st})} \quad (1)$$

$$\tau FAH_{st} = \frac{(FAH(\text{spending with taxes per capita})_{st} - FAH(\text{spending without taxes per capita}_{st}))}{FAH(\text{spending without taxes per capita}_{st})} \quad (2)$$

Much of the prior literature relies on the FoodAPS data from USDA, a single-round cross-sectional survey conducted between 2012 and 2013 covering 4,826 nationally representative households (United States Department of Agriculture Economic Research Service, 2026). The CPS data used in the present study offers several advantages over FoodAPS. First, the substantially larger sample size enhances statistical precision, particularly under the assumption of heteroskedasticity. Second, and more critically, the cross-sectional nature of FoodAPS precludes the identification of state-level and year-specific variation, making it impossible to isolate temporal and spatial effects on food expenditure. The CPS data, by contrast, records both the state and year of each observation, enabling a panel-based empirical strategy that distinguishes time and geographic sources of variation in food spending.

The analysis is restricted to the panel data from 2020 to 2024 to ensure longitudinal consistency in the data structure. The COVID-19 pandemic and the associated federal lockdown measures precipitated a structural shift in food consumption behaviors since 2020. Spending at quick service restaurants is slightly lower compared to 2019, while spending at full-service restaurants is 20% down in 2022. Subsequent year-over-year comparisons indicate that the pattern of lower FAFH sales and higher FAH sales relative to 2019 persisted consistently, suggesting a lasting structural realignment in food consumption rather than a transitory shock (Marchesi and McLaughlin, 2024). The pandemic also induced structural shifts in

broader demographic variables, most notably elevated unemployment among certain population groups (Antipova, 2021). By restricting the sample to observations from 2020 onwards, the analysis isolates food expenditure patterns from consumption behaviors after the pandemic which drifts structurally, thereby yielding more stable and internally consistent estimates.

As a survey instrument, the CPS contains observations with non-substantive response codes, including refusals, "don't know" responses, and non-responses. Only observations with valid values across all variables are retained, yielding a final analytical sample of 71,844 observations. Table 1 presents descriptive statistics for all variables included in the regression, together with category descriptions for each dummy variable. Control variables including a variety of individual level time varying factors, including age, education, children, employment and living status, current marriage status, disability status and if the income is below 185% poverty standard. As the summary statistics indicate, FAFH tax rates are, on average, substantially higher than FAH tax rates. However, the FAH tax rate exhibits considerably greater volatility across years and states, evidenced by a higher standard deviation relative to its narrower data range, reflecting the significant cross-state variation in grocery tax policy. The sample displays notable asymmetries across demographic characteristics. categories under each dummy variable. The sample is predominantly composed of respondents without a college degree (83.0%), and unmarried individuals account for approximately 71% of observations. The vast majority of respondents are US citizens (97.0%), and most report being outside the labor force (99.5%). By contrast, several other characteristics exhibit a more balanced distribution across the sample. For instance, recipients and non-recipients of SNAP benefits are represented in approximately equal proportions, at 50.6% and 49.4% respectively, which provides a favorable distributional basis for estimating heterogeneous tax effects across benefit status.

Table 1: Summary Statistics (N=71,844)

Variables	N	Mean / Prop.	Std. Dev.	Min	Max
Continuous Variables					
Age	71,844	53.208	14.104	15.00	85.00
Children	71,844	0.202	0.639	0.00	7.00
Food Expenditure Last Week ($)	71,844	169.489	120.958	10.00	600.00
Log Food Expenditure	71,844	4.870	0.768	2.30	6.40
FAFH Tax Rate	71,844	12.677	2.377	5.22	19.42
FAH Tax Rate	71,844	2.804	2.699	0.00	9.52
Categorical Variables					
Education	71,844	—	—	—	—
No College Degree	**59,664**	**0.830**	—	—	—
College Degree	12,180	0.170	—	—	—
Marriage	71,844	—	—	—	—
Married	21,084	0.293	—	—	—
Not Married	50,760	0.707	—	—	—
Disability	71,844	—	—	—	—
Yes	67,368	0.938	—	—	—
No	4,476	0.062	—	—	—
Employment	71,844	—	—	—	—
Employed	120	0.002	—	—	—
Unemployed	228	0.003	—	—	—
Not in Labor Force	71,496	0.995	—	—	—
Living Quarter Status	71,844	—	—	—	—
Owned	38,856	0.541	—	—	—
Rented	31,404	0.437	—	—	—
Occupied	1,584	0.022	—	—	—
Metropolitan Status	71,844	—	—	—	—
Metro	49,380	0.687	—	—	—
Non-Metro	22,464	0.313	—	—	—
Citizenship	71,844	—	—	—	—
Non-Citizen	2,172	0.030	—	—	—
Citizen	69,672	0.970	—	—	—
Benefits	71,844	—	—	—	—
Yes	36,372	0.506	—	—	—
No	35,472	0.494	—	—	—
Poverty	71,844	—	—	—	—
Below 185% Poverty	57,792	0.804	—	—	—
Above 185% Poverty	14,052	0.196	—	—	—

4. Methods

We use a two-way fixed effect model to assess the causal effect of FAFH tax rates on food expenditure. To accurately capture the variation and compress the heteroskedasticity, we use the logarithmic transformation of weekly grocery expenditure as the response variable. The main explanatory variable in the regression is the FAFH tax rate, calculated as per the equation mentioned in the data section. We utilize controlled variables including FAH tax rates, SNAP participation, age, sex, race, children, college degree, marriage, disability, employment, citizenship, valid address, living in metropolitan area or not, and if the household income is above 185% of the poverty line.

Table 2: Distribution of FAFH Tax Rates, By Interval

Tax Rate Interval	**N**	**Proportion (%)**
5% - <7%	2,460	3.42%
7% - <9%	1,080	1.50%
9% - <11%	14,376	20.01%
11% - <13%	18,324	25.51%
13% - <15%	26,628	37.06%
15% - <17%	5,736	7.98%
17% - <19%	2,940	4.09%
19% - 20%	300	0.42%
Total	**71,844**	**100.00%**

Min: 5.2173% | Max: 19.4192% | Mean: 12.6771% | Median: 12.9530% | Std Dev: 2.3770%

Table 2 presents the distribution of FAFH tax rates across the 71,844 observations in the sample, grouped into eight intervals ranging from 5% to 20%. The distribution is right-skewed and heavily concentrated in the middle-to-upper range. The three intervals from 9% to 15% together account for approximately 82.6% of all observations (20.01% + 25.51% + 37.06%), with the single largest mass sitting in the 13%-15% interval at 37.06%. The tails are thin: fewer than 5% of observations fall below 9%, and fewer than 5% fall above 17%. The gap between the two lowest intervals (5%-7% at 3.42% and 7%-9% at 1.50%) and the dense middle cluster is particularly stark, reflecting the discrete, policy-driven nature of state-level FAFH tax setting rather than a smooth continuous distribution. With only

4.92% of observations falling below 9% and the bulk of variation occurring between 9% and 15%, the effective identifying variation in the FAFH tax rate is relatively narrow. This concentration makes it harder for the regression to detect modest tax effects. The estimated results should be interpreted as reflecting variation primarily within the 9-15% range rather than across the full theoretical tax spectrum.

Table 3: Distribution of FAH Tax Rates, By Interval

Tax Rate Interval	**N**	**Proportion (%)**
0% - <1%	21,984	30.60%
1% - <2%	18,072	25.15%
2% - <3%	4,572	6.36%
3% - <4%	3,960	5.51%
4% - <5%	7,404	10.31%
5% - <6%	4,224	5.88%
6% - <7%	2,292	3.19%
7% - <8%	4,992	6.95%
8% - 10%	4,344	6.05%
Total	**71,844**	**100.00%**

Min: 0.0000% | Max: 9.5200% | Mean: 2.8041% | Median: 1.7606% | Std Dev: 2.6994%
Note: Most states exempt groceries from sales tax, consistent with low FAH tax rates.

Table 3 describes the distribution of FAH tax rates across the same sampling data, grouped into nine intervals from 0% to 10%. The distribution is right skewed as well, but mostly clustered at the lower end. The two lowest intervals, 0%-1% and 1%-2%, together account for 55.75% of all observations, meaning more than half the sample faces a grocery tax rate below 2%. The 0%-1% interval alone contains 30.60% of observations, reflecting the widespread practice among US states of exempting groceries from sales tax entirely or applying only nominal rates. Observations thin out considerably above 3%, with no single interval above that threshold exceeding 10.31% of the sample. The upper tail is sparse: only about 6% of observations face rates between 8% and 10%.

FAFH rates are on average more than four times higher than FAH rates, consistent with the longstanding policy norm in the United States of taxing restaurant food more heavily than grocery food, often as a de facto luxury or convenience tax. Besides, despite having a nar-

rower standard deviation in absolute terms, FAFH rates are actually more tightly clustered relative to their mean (coefficient of variation of 18.8%) than FAH rates (coefficient of variation of 96.3%), indicating that grocery tax policy is far more dispersed and heterogeneous across states than restaurant tax policy. This greater dispersion in FAH rates is driven by the bimodal policy landscape where many states impose zero grocery tax, while others apply moderate to substantial rates. This is shown by a sharply skewed distribution, as seen in Table 3. The floor of the FAH distribution is zero, whereas the FAFH distribution has a floor of 5.22%, confirming that no state in the sample fully exempts restaurant food from taxation, whereas many do so for groceries.

Economic data is systematically shaped by cyclical macroeconomic shocks and geographic factors, including state-level structural characteristics, price inflation, and broader macroeconomic cycles. Figures 1, 2, and 3 illustrate the distribution of weekly grocery expenditure across years, states, and FAFH tax rate intervals, respectively, and collectively motivate the empirical design choices described below. To account for both time-invariant state-level heterogeneity and common year-specific shocks, a two-way fixed effects (TWFE) linear regression model is employed as the primary specification.

Figure 1 presents a box plot overlaid with individual data points, illustrating the distribution of weekly grocery expenditure across all states from 2020 to 2024. The median expenditure in 2020 stands at approximately \$110-\$115, rising to around \$130-\$140 by 2021 and remaining at a similar level in 2022, before increasing more sharply to approximately \$150 in 2023 and stabilizing at a comparable level in 2024. The most pronounced structural change occurs between 2021 and 2022, where the interquartile range widens dramatically, approximately doubling in height, indicating a substantial increase in spending dispersion within-year. This elevated dispersion persists throughout 2023 and 2024. The lower whisker holds near \$10 in most years, reflecting a stable floor for minimum reported expenditure. The upper whisker reaches its maximum extension in 2022 at approximately \$500-\$550, indicating a greater

prevalence of high-spending outliers that year, with similarly elevated upper bounds in 2023 and 2024 relative to 2020 and 2021.

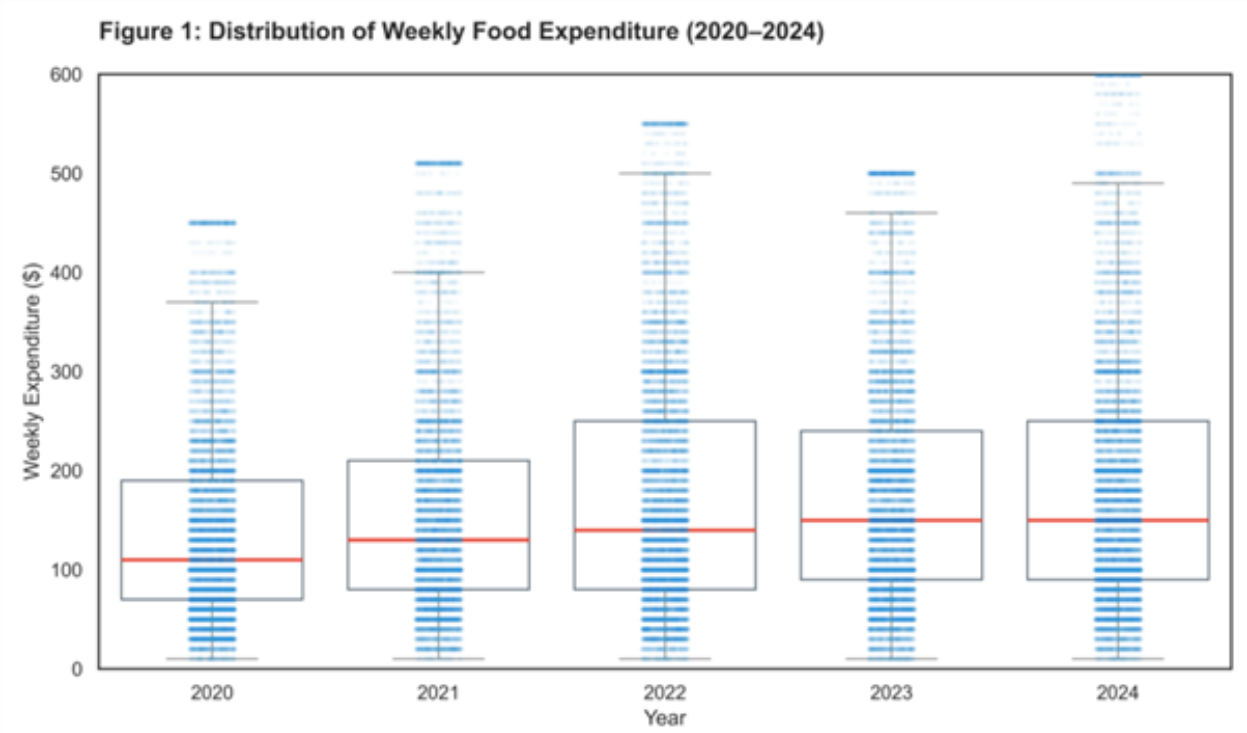


Figure 1: Distribution of Weekly Grocery Expenditure (USD) Over Time

The jitter overlay confirms that observations are densely concentrated in the lower half of the distribution in all years, consistent with the pronounced right skew visible across the full sample period. Taken together, Figure 1 points to a broad upward shift in the level and dispersion of grocery expenditure over the panel, motivating the use of log transformation and robust standard errors in the regression specification.

Figure 2 presents the cross-state distribution of median weekly grocery expenditure. The red dotted line along with the bar inside the boxes suggests that most state medians cluster tightly around the $140 overall median, suggesting broadly similar typical spending levels nationally. However, a few states such as Alaska, Florida, Iowa, and Mississippi have median values that deviate noticeably from the national median. Such variation likely reflects structural differences in cost of living associated with geographic location. For instance, Alaska

consistently ranks among the highest-cost states in the United States, which may contribute to its elevated grocery expenditure. It should be noted, however, that the exclusion of observations with missing or invalid food expenditure values may introduce a degree of sample selection bias into state-level comparisons.

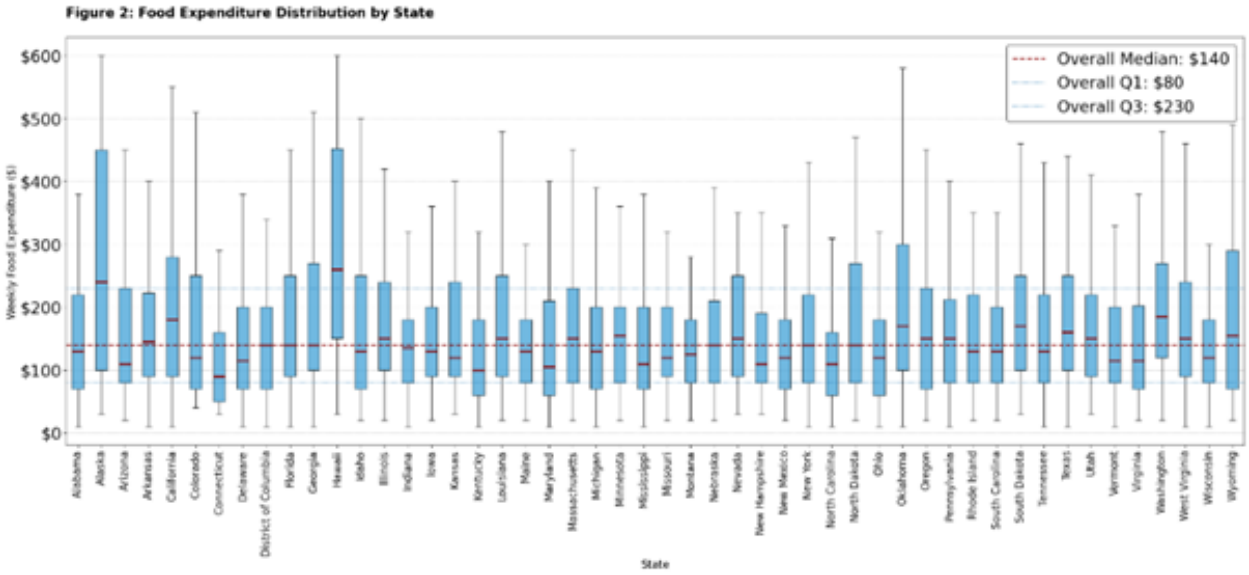


Figure 2: Distribution of Median Weekly Grocery Expenditure (USD) by State
Note: (1) The red line within the boxes indicates the state median. The boxes suggest a 25% and 75% quantile of the expenditure in each state. The whiskers extend to the 1.5 times quantile range from each quartile. (2) The states are arranged by state codes, which is coded by the alphabetical sequence of the total 51 states (Washington D.C. included) in the United States.

Figure 3 presents a scatter plot of weekly grocery expenditure against the primary explanatory variable, the FAFH tax rate. Observations are concentrated within two distinct tax rate intervals: 5-8% and 9-20%. The lower interval contains a relatively sparse cluster, while the higher interval hosts the bulk of observations, reflecting the distributional pattern documented in Table 2. A notable gap exists between 8% and 9%, consistent with the discrete variation in state tax policy. Both clusters display a pronounced upper tail, indicative of high-spending outliers. Critically, no systematic trend in grocery expenditure is visually discernible across the range of FAFH tax rates, suggesting a weak unconditional correlation between the dependent variable and the main regressor. This observation underscores the importance of controlling for confounding household and geographic characteristics in the

regression framework, as the raw bivariate relationship offers limited signal.

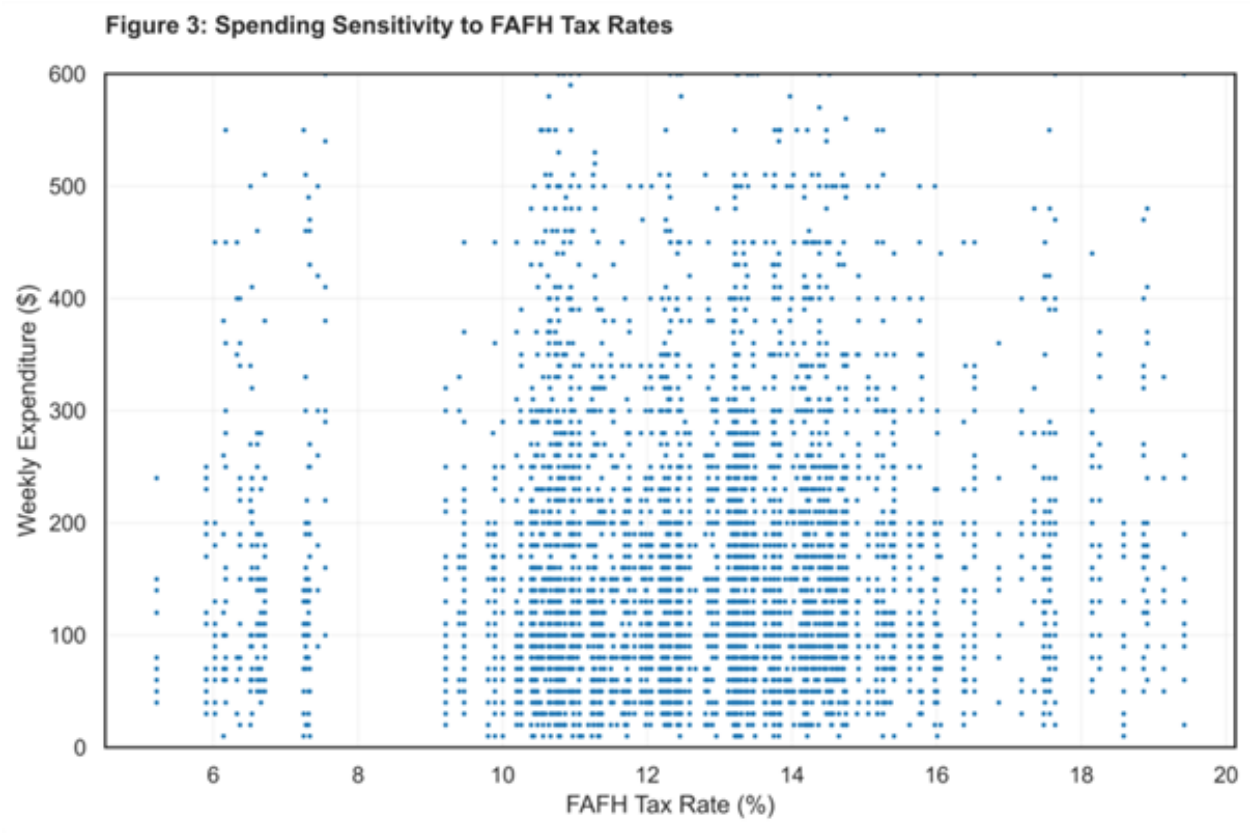


Figure 3: Weekly Grocery Expenditure (USD) vs FAFH Tax Rates

The distributional patterns observed across Figures 1 to 3 inform three key decisions in the empirical strategy. First, the pronounced right skew and concentration of high-spending outliers evident in Figures 1 and 3 motivate the use of the natural logarithm of weekly grocery expenditure as the dependent variable. Log transformation compresses the upper tail, reduces the undue influence of extreme values on the regression estimates, and yields coefficients that are interpretable as approximate percentage changes. It provides a more economically meaningful and robust representation of the tax-expenditure relationship than what a levels specification would do. Second, the distributions shown in Figures 2 and 3 are consistent with heteroskedasticity: spending dispersion varies markedly across states and tax rate intervals, indicating that the variance of the error term is unlikely to be constant across observations. To address this, heteroskedasticity-robust standard errors are computed, en-

suring that statistical inference remains valid even when the homoskedasticity assumption is violated. This approach complements the log transformation, which itself partially stabilizes variance across the distribution. Third, Figures 1 and 2 together reveal systematic divergence in grocery expenditure both over time and across states, consistent with the presence of time and geographic fixed effects. Left uncontrolled, these factors could confound the estimated relationship between FAFH tax rates and FAH spending by attributing variation arising from, for example, pandemic-era inflation or persistent state-level cost-of-living differences to the tax variable. A two-way fixed effects specification is therefore employed, incorporating state fixed effects to absorb permanent cross-state differences such as food culture, policy environments, and economic structure, and year fixed effects, to control for economy-wide shocks common to all states in a given year.

Following prior literature, we also include a rich set of demographic control variables to isolate the causal effect of the FAFH tax rate on grocery expenditure from confounding household characteristics, which cannot be absorbed by time and spatial fixed effects. The baseline regression equation is specified as follows:

$$\ln(realFAH_{pcst}) = \beta_1 \tau_{FAFH,st} + \beta_{controls,st} + \gamma_s + \lambda_t + \varepsilon_{st} \tag{1}$$

where $\ln(realFAH_{pcst})$ denotes the logarithm of household food-at-home spending in state s, year t. $\tau_{FAFH,st}$ denotes the state-level FAFH sales tax rate, which is the primary explanatory variable of interest. $Controls_{st}$ represents the vector of household-level demographic control variables, as detailed in Table 1, capturing individual-level heterogeneity that the state and year fixed effects cannot absorb. Finally, γ_s suggests the state fixed effect for state s (permanent cross-state differences), and λ_t suggests the year fixed effect for year t (common economy-wide shocks). ε_{st} represents the random error term.

As the median income does not vary much across the short time period of the dataset, it

is reasonable to take an assumption that the cyclical shocks affect all states identically. We therefore implement a one-way fixed effect model based on this assumption for comparison with the model (1). The model is described below:

$$\ln(realFAH_{pcst}) = \beta_1 \tau_{FAFH,st} + \beta_2 controls_{st} + \gamma_s + \varepsilon_{st} \tag{2}$$

The notation in model (2) is the same as model (1), except deleting the year fixed effects. The model is constructed mainly as a robustness check for model (1), which is more valid by intuition.

State-clustered standard errors are employed to account for residual within-state correlation in the error term across households. Even after conditioning on state fixed effects, households within the same state are likely to share unobserved local factors such as regional economic conditions, food retail market structure, and state-specific policy variation that induce correlated disturbances within clusters. With 51 state clusters (including Washington D. C.), the asymptotic properties of the cluster-robust variance estimator are well-supported. Clustering along the time dimension is omitted, as the panel contains only five years, yielding too few time clusters for the cluster-robust estimator to perform reliably.

5. Results

Table 4 describes the main regression results based on model (1) and (2) for comparison. The column (1) gives all estimated coefficients based on model (1), while the column (2) gives the results based on model (2). There are certain demographic characteristics that show discrete impact over one's weekly grocery expenditure. However, the result associated with the main explanatory variable suggests limited impact on grocery expenditure.

Model (1) yields a negative but statistically insignificant coefficient on the FAFH tax rate.

Table 4: Main Results of Regression (N=71,844)

Variable	**(1)**	**(2)**
Intercept	5.7377***	4.4707***
	(0.3067)	(0.2857)
FAFH Tax Rate	-0.0045	0.0989***
	(0.0215)	(0.0178)
FAH Tax Rate	0.0371**	0.0198
	(0.0150)	(0.0163)
SNAP Benefits	0.0307	0.0276
	(0.0269)	(0.0267)
Age	-0.0085***	-0.0084***
	(0.0007)	(0.0007)
Children	0.1620***	0.1616***
	(0.0129)	(0.0124)
Education		
College Degree	0.0039	0.0082
	(0.0181)	(0.0179)
Marriage		
Single	-0.2146***	
-0.2164***		
	(0.0252)	(0.0248)
Disability		
Not Disabled	0.0211	0.0252
	(0.0457)	(0.0466)
Employment		
Unemployed	-0.3221	-0.3390
	(0.2098)	(0.2161)
Not in labor force	-0.2502	-0.2511
	(0.1582)	(0.1559)
Living Status		
Rented	-0.1236***	-0.1252***
	(0.0265)	(0.0268)
Occupied	-0.1573*	-0.1531*
	(0.0828)	(0.0837)
Living in Metropolitan Area		
Non-metro Area	-0.0293	-0.0346
	(0.0357)	(0.0355)
Citizenship		
Citizen	-0.0855	-0.0781
	(0.0877)	(0.0856)
Poverty		
Income Above 185 Percent Poverty	0.1917***	0.1953***
	(0.0289)	(0.0290)
Adj R^2	0.0965	0.1042
N	71,844	71,844

*** p<0.01, ** p<0.05, * p<0.10. SE in parentheses, clustered by state.

Notes: Base categories: Education = No College Degree, Marriage = Married, Disability = Has Disability, Employment = Employed, Living Quarter = Owned, Metro = Metro Area, Citizenship = Non-Citizen, Poverty = Below 185% FPL.

After absorbing both time and state fixed effects, a one percentage point increase in the FAFH tax rate is associated with a 0.45 percent decrease in weekly grocery food expenditure. The estimate falls well short of conventional significance thresholds and cannot be distinguished from zero. Conditional on state-level structural differences and common time trends, the FAFH tax rate does not exert a statistically discernible independent effect on grocery food expenditure in this sample. In contrast, the positive and highly significant coefficient on FAFH tax rate under model (2) suggests that one percent of FAFH tax rate increase is associated with 9.89 percent increase in food expenditure, which is statistically significant in the significance level of 0.01. However, the result in model (2) is counterintuitive as grocery food is not likely to be inferior good. Under the model with time effect not absorbed, the estimated effect is more likely to be driven by the upward time trend. Both FAFH tax rates and food expenditure trended upward over 2020-2024 due to inflation and post-pandemic fiscal adjustments, creating a positive correlation that is not causal. Overall, FAFH tax rates have no statistically distinguishable effect on log food expenditure in this sample. The conclusion is weak considering the limitation in our tax rate measurements, which is discussed in the limitation section later.

The positive and significant coefficient in the Two-Way FE model suggests that each percent increase in FAH tax rates is associated with a 3.71 percent increase in grocery food expenditure. The coefficient is statistically significant at the significance level of 0.05. The finding is counterintuitive at first glance, as standard demand theory would predict that higher taxes on grocery food raise its effective price and reduce consumption. However, considering household budget constraints and food substitution patterns, for low-income households with income under the poverty thresholds, grocery food represents the primary and often only viable means of food consumption. Unlike higher-income households that maintain the financial flexibility to substitute between FAH and FAFH in response to price changes, budget-constrained households have limited scope to absorb tax-driven price increases through cross-category substitutions. Restaurant and food-away-from-home options

typically carry a substantially higher per-meal cost even before tax considerations, meaning that switching away from grocery food in response to a FAH tax increase is not a financially feasible adjustment for this segment of the population. As a result, lower-income households may respond to higher grocery taxes not by reducing food expenditure, but by maintaining or even increasing nominal spending to sustain adequate nutrition intake, which demonstrates inelastic or near-inelastic demand with respect to the FAH tax rate. This interpretation is consistent with the broader literature on food demand among low-income populations, which consistently finds that basic grocery expenditure is among the least price-responsive components of household budgets when no affordable substitutes are available.

SNAP receipt has no statistically significant effect on log food expenditure in either specification. From table 4 we can infer that several demographic characteristics, including age, number of children, marriage, address status, and household income situation show statistically significant impact on grocery expenditure.

Overall, the estimated results collectively suggest that FAFH taxation exerts limited and statistically indistinguishable effects on household grocery expenditure, while a broader range of demographic characteristics as well as FAH tax rates emerge as more consistent and economically meaningful determinants of weekly grocery spending. These findings carry several actionable implications for food tax policy design. From a fiscal policy perspective, taxation targeted at FAFH consumption, which are principally restaurant meals and prepared food services, represents a relatively efficient revenue-raising instrument with limited distributional distortion to essential grocery spending. As the evidence suggests, households operating under limited food budget constraints overwhelmingly depend on grocery food as their primary source of nutrition and exhibit limited price-driven substitution toward restaurant dining even as FAFH taxes rise. This structural inelasticity implies that FAFH tax increases are unlikely to substantially displace spending away from restaurants among the broader population, while simultaneously generating fiscal revenues that can be

directed toward welfare-enhancing programs. In this sense, FAFH taxation aligns reasonably well with the principles of progressive fiscal design imposing a relatively greater burden on higher-income households for whom restaurant consumption constitutes a larger share of the food budget, while leaving the grocery expenditure of lower-income households largely undisturbed.

To further strengthen the progressive character of food taxation and meaningfully improve the material living standards of economically vulnerable households, policymakers could consider a coordinated dual approach: gradually increasing FAFH tax rates to expand the fiscal base, while simultaneously reducing food-at-home tax rates or extending direct subsidies to grocery food suppliers to lower the effective cost of essential nutrition at the point of purchase. Such a policy rebalancing would redistribute the tax burden away from basic food necessities and toward discretionary food consumption, with potentially meaningful welfare gains for food-insecure and low-income populations. However, the null result on the estimated effect of FAFH tax rate in model (1) specification should not be interpreted as definitive evidence that taxation has no effect on food expenditure. The coefficient is better understood as a conservative lower bound whose precision is fundamentally constrained by the measurement limitations inherent in the constructed tax rate variable, particularly the bias introduced by aggregating across heterogeneous subsidy recipients and the absence of individual-level tax exposure variation. Future research should seek to replicate and extend this analysis using more precisely measured effective tax rate instruments, ideally constructed at the individual or household level to capture meaningful variation in actual tax burdens. Such improvements in measurement would substantially sharpen the identification of food tax elasticities and provide a more reliable empirical foundation for evidence-based food taxation policy.

6. Limitations

Our analysis has several limitations which weaken the estimated results due to the data structure. Such flaws include the designed structure and the increasing non-response rate after COVID-19 shock in CPS data, the construction of food tax rates, as well as the unresolved endogenous problem due to the availability of SNAP policy data.

First, there are several flaws in CPS survey data. The survey-based nature of the data introduces cognitive bias in self-reported expenditure. CPS data is collected through structured questionnaires, meaning that all expenditure figures reflect respondents' subjective recall and estimation rather than objective transaction records. As food expenditure is often distributed across multiple shopping trips, outlets, and payment methods within a week, respondents can be unfamiliar with the precise details of their household food spending. Therefore, their reported figures may drift systematically from actual expenditure levels. Such bias introduces heterogeneous measurement error across the sample that standard econometric techniques cannot fully correct. The data also fails to account for diet-relative preferences at individual level, which are used in some of the research and account for variation in people's food consumption behaviors. The absence of such preference indicators in the CPS data means that a potentially important source of individual-level variation in food spending is omitted from the analysis, which may bias coefficient estimates on other regressors that are correlated with dietary preferences. The precision of the expenditure measure is further constrained by the rounding convention applied in the survey instrument. After recording the estimated grocery expenditure from respondents, the answers are automatically rounded to the nearest ten dollars, which imposes a discrete, coarsened structure on what is inherently a continuous variable. This artificial rounding reduces measurement precision, compresses the within-individual variation in reported expenditure, and may introduce systematic heaping in the distribution. In combination with the log transformation applied in the regression, this rounding may subtly distort the distributional properties of the dependent variable,

particularly in the lower expenditure range where a ten-dollar rounding increment represents a proportionally larger measurement error. Some people also argue CPS data is not a valid representation of the population after the pandemic. Ward and Edwards found that the response rate has been declining after the COVID-19 shock. In the first four months of the pandemic period (March-June 2020), the average month-over-month nonresponse rate increased by 58 percent, while the size of newly entering cohorts declined by 37 percent compared to the prior 15 months. There is also a wide gap between the estimates of unemployment rate from CPS data and other sources, which raises a concern for not representing the actual demographic characteristics on CPS data. (Ward & Edwards, 2021)

Second, our tax rate measure lacks precise variation due to the construction methodology. Rather than drawing on statutory tax rates stipulated by state policy, the FAFH tax rate is calculated by comparing food sales revenues with and without tax incidence. This approach introduces systematic downward bias in the estimated tax rate. Because households participate in SNAP or receive food-related subsidies, their effective tax burden is partially or fully offset by program benefits. SNAP benefits are explicitly designed to reduce out-of-pocket food expenditure, meaning that purchases made through SNAP are effectively untaxed at the point of sale. When these transactions are aggregated into the state-level sales data used to back-calculate tax rates, the resulting measure conflates the statutory tax rate with the distribution of subsidy receipt across the population. Consequently, the calculated tax rate systematically understates the true prevailing FAFH tax level, and the degree of understatement varies across states in proportion to their SNAP participation rates. Therefore, the measured tax rate is a noise-contaminated proxy for the true rate, which mechanically biases the coefficient estimate toward zero and reduces statistical power. The rates applied to each respondents lack individual-level variation by construction also, as the calculated tax rate is a state-level aggregate measure that is assigned uniformly to all respondents within the same state, regardless of their actual tax exposure and subsidy status. The assignment strategy implies that the tax rate variable contains no within-state, cross-individual variation, where

every resident of a given state receives an identical value in every year. It is therefore hard to distinguish the tax rates from a state-level fixed characteristic, as it is largely collinear with state fixed effects as discussed earlier. The individual heterogeneity in effective tax burden driven by differences in subsidy receipt, purchasing channel, and consumption basket composition, is also entirely suppressed. Third, it implies that standard errors clustered at the state level may understate the true sampling uncertainty, as the identifying variation in the key regressor is effectively reduced to the number of distinct states rather than the full individual-level sample size, substantially reducing the degrees of freedom available for inference.

We are also unable to resolve the endogenous problem of the SNAP participation indicator due to the limits on data availability. As our data is panel, we apply the time and spatial fixed effect to absorb variation caused by economic cycles and time-invariant within-state characteristics. Following the research of Ratcliffe, McKernan, and Zhang (2011), as well as Zheng et al. (2019), we consider three policy indicators illustrating the eligibility requirements and administrative patterns of the SNAP program in the state level: if the state uses fingerprint scan for identity verification, if the state adopts simplified reporting, and if the local SNAP program enables outreach spending (Zheng et al., 2019). We extract data for the instrumental variables from USDA's SNAP policy data sets, with the latest data available for outreach spending in September 2020, and for adoption of simplified report in Feb 2020. As Zheng et al. illustrate, the relevant policy changes do not occur frequently over time (Zheng et al., 2019). We therefore expect no change in the SNAP administrative process and the eligibility requirements from 2020 to 2024, and thus assign the latest available value of the instrumental variables based on the observations' state code. The distribution of three variables is shown in table 6.

Since the variables themselves are state-level data, there is no variation within the state. Similarly, since we must assume no changes in the value of these variables in 2020-2024,

Table 5: Distribution of Potential Instrumental Variables (N=71,844)

Variables	N	Mean / Prop.	Std. Dev.	Min	Max
Outreach	71,844	313.465	664.646	0.00	2664.72
Reportsimple	**71,844**	—	—	—	—
No	4,020	0.056	—	—	—
Yes	67,824	0.944	—	—	—
Fingerprint	**71,844**	—	—	—	—
No Fingerprint	71,844	1.000	—	—	—
With Fingerprint	0	0.000	—	—	—

Note: Add notes here

there is no change across time. Therefore, the variation in the instruments is fully absorbed in the two-way fixed effect model applied in the analysis and does not possess econometric meaning on regressing them on the SNAP participation.

7. Conclusion

In this paper, we aim to address the effect of FAFH tax rates on household grocery expenditure, drawing on a panel of 71,844 observations from the Current Population Survey (CPS) and the USDA food tax database spanning 2020 to 2024. Employing a two-way fixed effects regression with state-clustered heteroskedastic standard errors, the analysis seeks to isolate the causal relationship between FAFH taxation and FAH spending while controlling for time and spatial fixed effect, as well as a comprehensive set of demographic characteristics.

The central finding of the analysis is that FAFH tax rates do not exert a statistically significant effect on household grocery expenditure once both state and year fixed effects are incorporated into the model. The estimated coefficient on the FAFH tax rate under the baseline two-way fixed effects specification is negative but indistinguishable from zero, which implies that conditional on time-invariant geographic differences and common annual shocks, increases in restaurant food taxation do not meaningfully redirect household spending toward or away from grocery food. The FAH tax rate is found to be positively and significantly

associated with household grocery expenditure, a counterintuitive result that is most plausibly explained by near-inelastic demand for basic food necessities among budget-constrained households. For this segment of the population, grocery food constitutes the primary and often only financially viable source of nutrition, such that higher grocery taxes translate into increased nominal spending rather than reduced consumption or cross-category substitution toward restaurant food. SNAP participation does not attain statistical significance in either specification, while a range of demographic characteristics including age, number of children, marital status, employment status, racial identity, and income position relative to the federal poverty line,emerge as robust and economically meaningful determinants of weekly grocery expenditure.

From a policy standpoint, these findings suggest that FAFH taxation represents a relatively efficient and progressively aligned revenue instrument. Because FAFH taxes do not appear to substantially displace grocery spending among lower-income households, they can generate fiscal revenues without imposing additional burdens on essential food consumption. A coordinated dual-policy approach: gradually expanding FAFH tax rates while simultaneously reducing FAH tax rates or extending direct subsidies to grocery food suppliers, has the potential to redistribute the food tax burden away from necessities and toward discretionary food consumption, with potentially meaningful welfare gains for food-insecure and economically vulnerable households. However, several important limitations apply to the conclusions above. The tax rate is constructed from aggregate sales data rather than statutory rates, introducing systematic downward bias through the inclusion of subsidy-exempt SNAP transactions. This measurement error attenuates the estimated coefficient toward zero and reduces statistical power, meaning that the null result should be interpreted as a conservative lower bound on the true effect rather than definitive evidence of a zero impact. Additional limitations include cognitive bias and rounding error in self-reported CPS expenditure data, increasing survey non-response rates in the post-pandemic period, and the inability to resolve the endogeneity of SNAP participation using available state-level policy

instruments, which are fully absorbed by the two-way fixed effect structure.

In response to the limitations, future research should seek to address these limits through more precisely measured effective tax rate instruments, ideally constructed at the individual or household level to capture meaningful heterogeneity in actual tax burdens. Incorporating diet-related preference indicators and leveraging higher-frequency transaction-level data would further sharpen the identification of food tax elasticities and provide a more robust empirical foundation for evidence-based food taxation policy.

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